

CANDIDATE NAME _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 Preliminary Examination 2014

H1 BIOLOGY
8875

Paper 2

2 Hours

Additional materials:
Writing papers / OMS

Date / Day: Tuesday 19th August 2014**INSTRUCTIONS TO CANDIDATES**

Write your name, CG and index number in the spaces at the top of this page and on all separate writing papers used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Write your answers in spaces provided on the question paper.

Section B

Answer **only one** question out of two.

Write your answers on the separate answer paper provided.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Section A	
1	/8
2	/9
3	/8
4	/8
5	/7
Section B	
6	/20
7	/20
TOTAL	/60

This question paper consists of 14 printed pages

Question 1

Porins are transmembrane proteins found on the outer membrane of mitochondria. Their function is to transport substances across the membrane. **Fig. 1.1a** and **Fig. 1.1b** show the side view and top view of the structure of a single polypeptide of porin respectively.

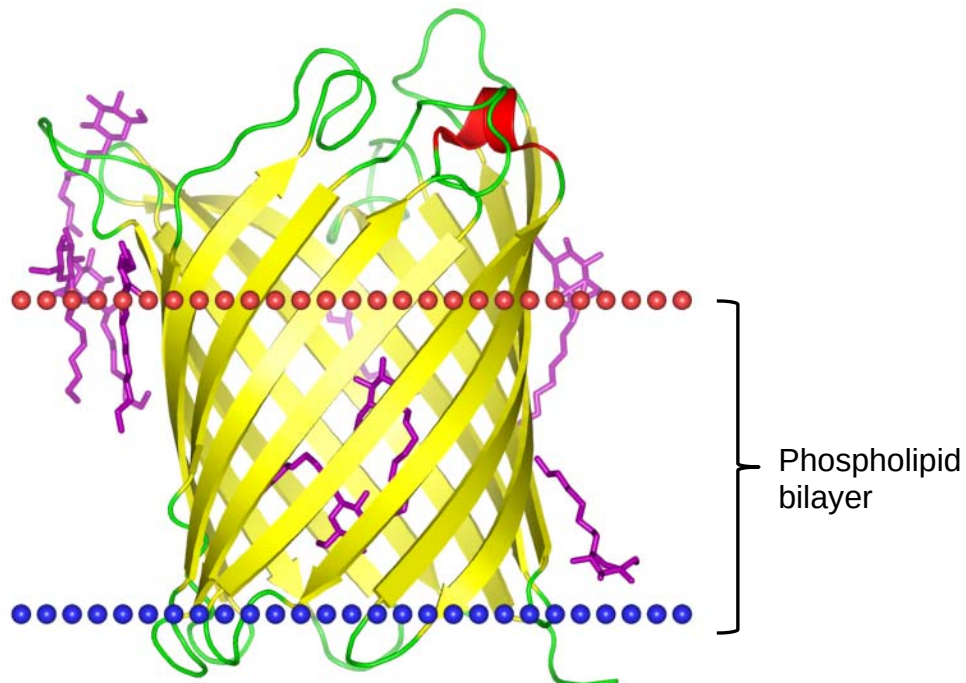


Fig. 1.1a



Fig. 1.1b

(a) Name a covalent bond present within the protein porin. [1]

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(b) With reference to **Fig. 1.1a** and **Fig. 1.1b**,

(i) Name a non-covalent bond present within the protein porin. [1]

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(ii) Describe the structure of the protein porin. [2]

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A researcher intends to study the transport of substances carried out by porins across the outer mitochondrial membrane. The researcher begins by isolating mitochondria from rat liver cells and placing them in 5 different solutions, each containing a different substance with a concentration of 10 arbitrary units as shown in **Table 1.2**.

Table 1.2

Solution	Content of Solution
1	10 AUs of Na ⁺ ions
2	10 AUs of Cl ⁻ ions
3	10 AUs of H ⁺ ions
4	10 AUs of protein insulin
5	10 AUs of acidic amino acid glutamate

The concentrations of the substances in the intermembrane space of the mitochondria were measured before the experiment and after 15 minutes of being placed in the solutions. The results obtained by the researcher are shown in **Table 1.3** on the following page (page 4).

Table 1.3

Solution	Content of Solution	Concentration of substances in mitochondrial intermembrane space before experiment	Concentration of substances in mitochondrial intermembrane space after 15 minutes
1	10 AUs of Na ⁺ ions	1 AU of Na ⁺ ions	10 AUs of Na ⁺ ions
2	10 AUs of Cl ⁻ ions	2 AUs of Cl ⁻ ions	10 AUs of Cl ⁻ ions
3	10 AUs of H ⁺ ions	2 AUs of H ⁺ ions	10 AUs of H ⁺ ions
4	10 AUs of protein insulin	0 AU of protein insulin	0 AU of protein insulin
5	10 AUs of acidic amino acid glutamate	0 AU of acidic amino acid glutamate	9 AUs of acidic amino acid glutamate

(c) With reference to **Table 1.2** and **Table 1.3**, suggest how the outer mitochondrial membrane is responsible for the concentrations of substances in mitochondrial intermembrane space after 15 minutes. [4]

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[Total: 8 marks]

Question 2

Scientists often employ the method of fluorescence imaging to study the process of nuclear division in cells. This method involves the use of fluorescence tagged antibodies to label specific structures within the cell, so that their movements and behaviours can be followed over the course of nuclear division.

A group of SR scientists decided to employ the aforementioned method to study the changes in centromere number in an animal cell undergoing meiosis. Initial testing showed that each centromere could be visualised as a single dot and therefore the total number of dots can be taken to correspond to the total number of centromeres within the cell. **Table 2.1** shows the number of dots and hence centromeres observed per cell, at various stages of meiosis.

Table 2.1

Stage of Meiosis	Prophase I	Prophase II	Anaphase II	Telophase II
Number of dots observed per cell	16			16

(a) With reference to **Table 2.1**,

- (i) Predict the number of dots (i.e. centromeres) observed in prophase II and anaphase II respectively. [2]

Prophase II:

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Anaphase II:

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- (ii) Explain your answer to part **(a) (i)**. [2]

Prophase II:

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Anaphase II:

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When the same experiment was conducted while the cell was in interphase, prior to meiosis I, no distinct dots could be observed in the nucleus. Instead, the entire nucleus emitted fluorescence.

(b) Suggest a reason for this observation. [1]

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Apart from using fluorescence imaging to determine the changes in centromere numbers, this technique was also employed by the SR scientists to study the movement of chromosomes during meiosis in the animal cell. More specifically, the changes in distance between 2 corresponding regions of non-sister chromatids of a pair of homologous chromosomes (**X**) were measured over the course of the first meiotic division. The results are shown in **Fig 2.1** below.

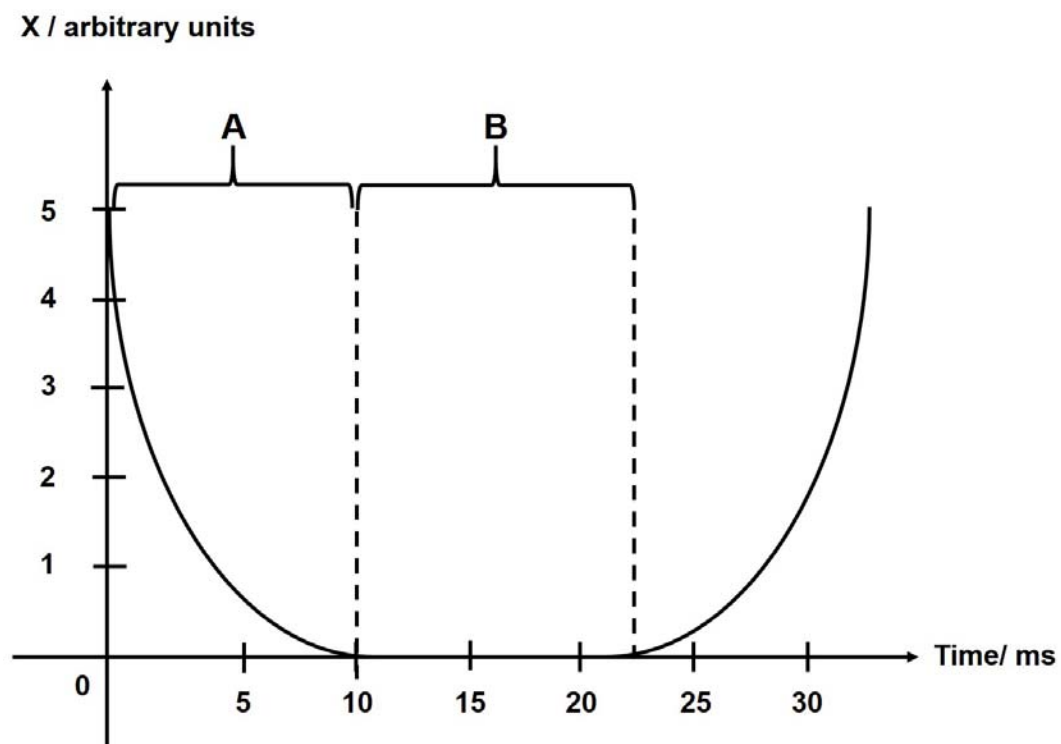


Fig 2.1

(c) With reference to **Fig 2.1**, explain the changes in **X** for

(i) Region A. [2]

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(ii) Region B. [2]

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[Total: 9 marks]

Question 3

In an experiment to study ATP synthesis in plants, chloroplasts are isolated, and the pH levels in the various compartments are monitored. The graph below shows the results of the experiment.

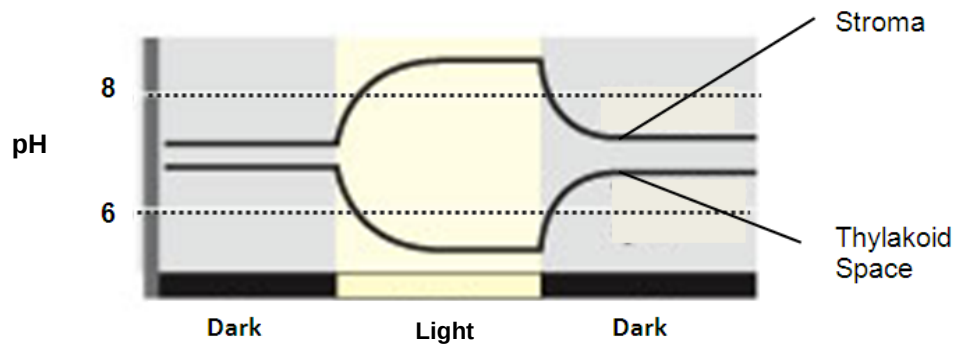


Fig 3.1

(a) Explain the changes in pH seen in **Fig 3.1** upon illumination. [3]

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(b) Describe a similarity and a difference in the production of ATP in chloroplasts and in mitochondria. [2]

Similarity:

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Difference:

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Fig 3.2 shows some of the reactions during the light-independent reactions of photosynthesis.

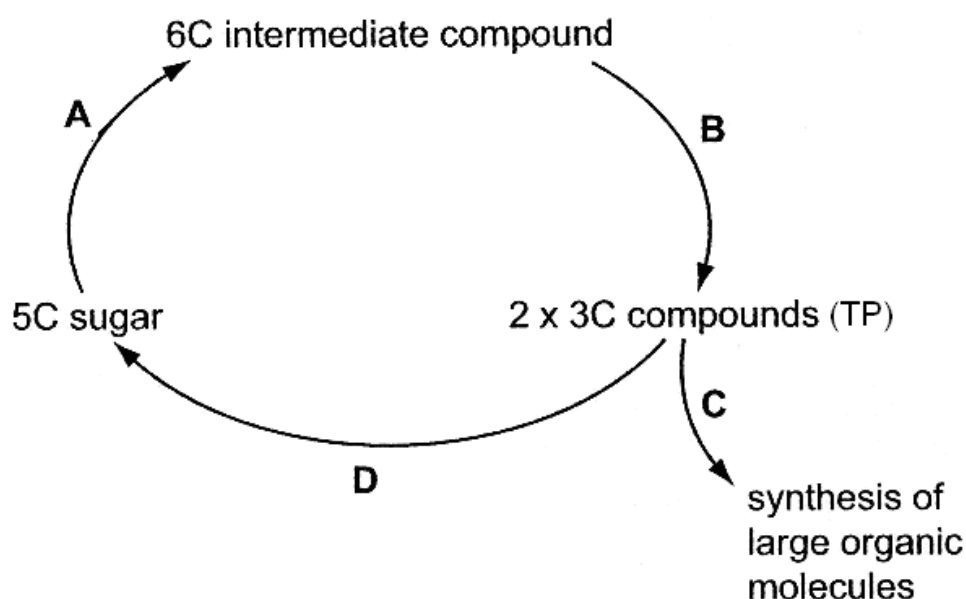


Fig 3.2

(c) With reference to **Fig 3.2**,

- (i) In which stage(s) will both products of the light-dependent reactions be used? [1]

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- (ii) Describe how the products of the light-dependent reactions are used in the light-independent stage of photosynthesis. [2]

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[Total: 8 marks]

Question 4

Adalia bipunctata, the 2-spot ladybug (or ladybird), is a small beetle which exists in two forms. The typical form has red wing cases with two black spots. The red colour is a signal for predators of its unpleasant taste. The other less common form, melanic, has black wing case with two red spots. **Fig 4.1** shows the 2 different forms of ladybugs.

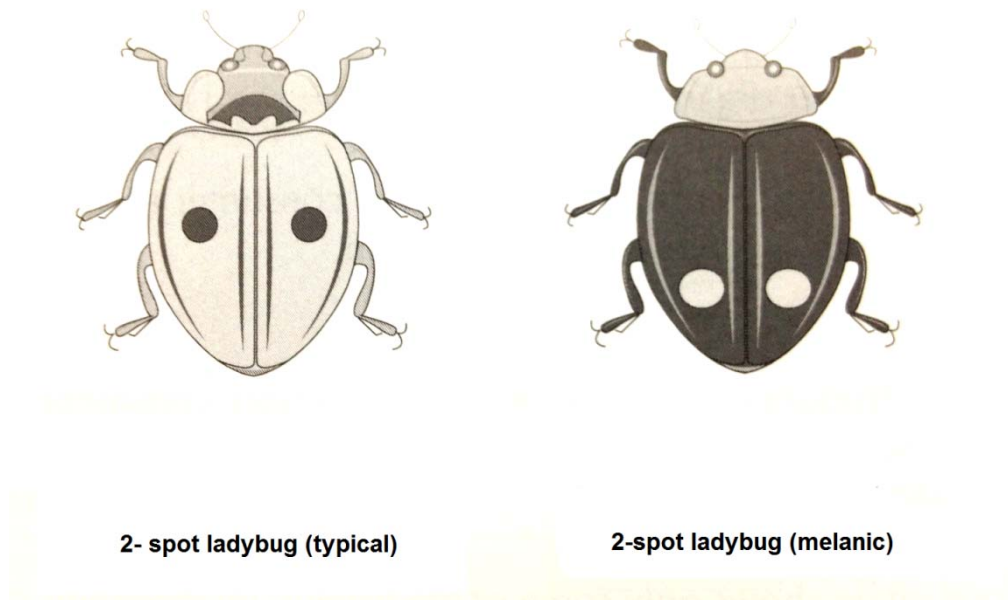


Fig 4.1

The melanic form used to be very common in industrial areas of Britain but declined after 1960. **Fig 4.2** shows the frequency of melanic ladybugs over the years and the correlation to smoke present in the air.

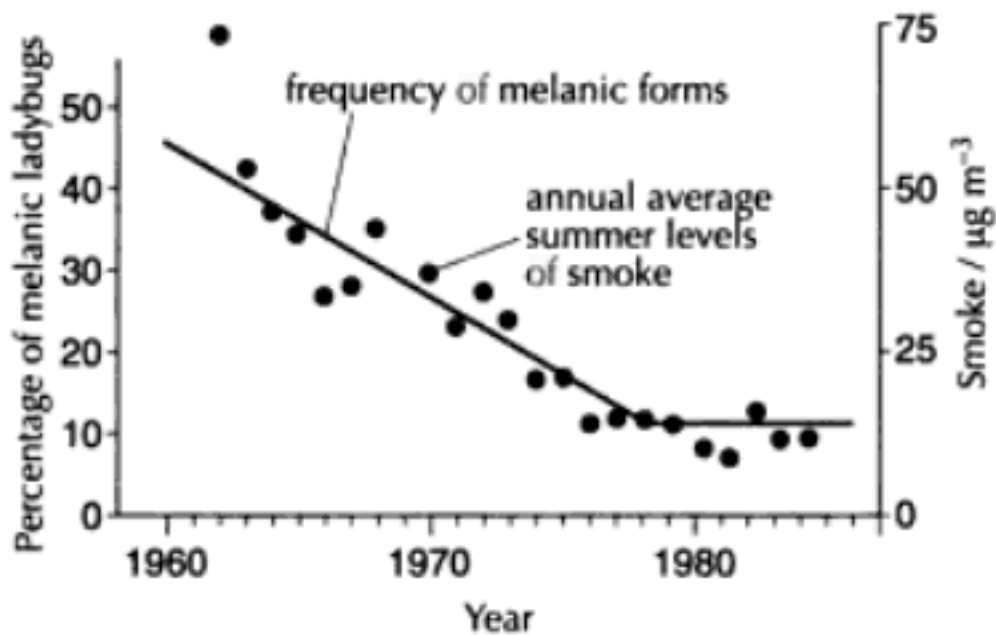


Fig 4.2

- (a) Using **Fig 4.2**, explain how the process of natural selection had led to the changes in the frequencies of the melanic ladybug over the years. [4]

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- (b) A region of DNA coding for NADH dehydrogenase subunit was isolated for analysis between the three species - *Adalia bipunctata* (2-spot ladybug), *Coccinella septempunctata* (7-spot ladybird) and *Propylea quattuordecimpunctata* (14-spot ladybird). Part of the DNA sequence coding for NADH dehydrogenase subunit found in these three species are shown in **Fig. 4.3**.

<i>Adalia bipunctata</i>	ACCAGCCTGTGCATCGATGACGACTAAGTGATACCATAAAGACT
<i>Coccinella septempunctata</i>	ACTAGCCTGTGCATCGATGACGACTAAGTGATACCATAAAGACT
<i>Propylea quattuordecimpunctata</i>	ACGAGCCTGTGCATCGATGACGACTAAGTGATACCATAAAGACT

Fig. 4.3

- (i) Suggest why NADH dehydrogenase gene is used for analysis. [1]

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- (ii) What is the term used to describe the relationship between the sequences shown in **Fig 4.3**? [1]

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- (iii) Explain how the relationship between the DNA sequences coding for NADH dehydrogenase subunit in **Fig. 4.3** provides evidence to support the theory of evolution. [2]

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[Total: 8 marks]

Question 5

A monkey homozygous for a mutation on the insulin gene suffers from Type 1 diabetes. An experiment utilizing genetic engineering techniques was conducted to obtain genetically modified stem cells in order to treat the disease.

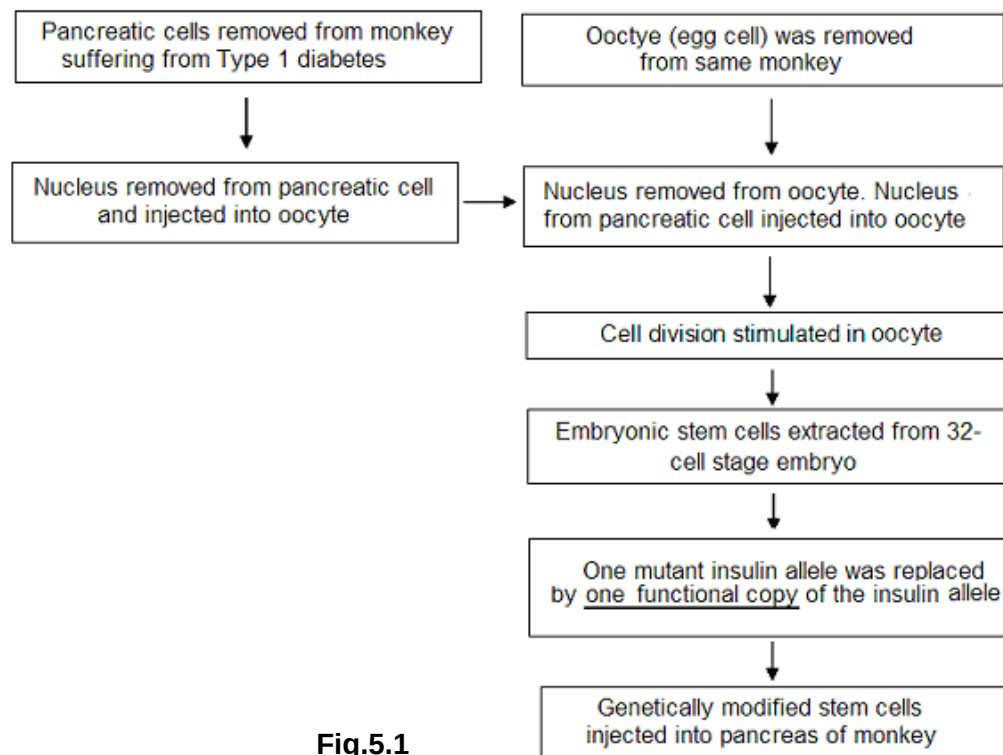


Fig.5.1

- (a) Describe the role of embryonic stem cells in a normally developing embryo. [2]

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(b) Compare between embryonic stem cells and adult stem cells. [2]

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(c) Suggest why it is sufficient to insert a single functional allele into the stem cell. [1]

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(d) Discuss **one** ethical concern regarding the use of embryonic stem cells. [1]

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[Total: 7 marks]

Section B (20 Marks)

Answer **one** question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in sections (a), (b), etc as indicated in the question.

Question 6

- (a) Explain how the structure of DNA differs from tRNA. [4]
- (b) Describe the polymerase chain reaction and contrast it with DNA replication. [10]
- (c) Explain, with an example, how a change in the sequence of the DNA nucleotide may affect the amino acid sequence in a protein and hence the phenotype of the organism. [6]

Question 7

- (a) Describe the main structural features and functions of endoplasmic reticulum, nucleolus and ribosome. [10]
- (b) Explain how gel electrophoresis is used to analyse nucleic acids. [6]
- (b) Discuss the problems associated with the expression of eukaryotic genes in prokaryotes and suggest how these problems can be overcome. [4]